

# **Institute of Actuaries of India**

## **Subject SP7 – General Insurance Reserving and Capital Modeling**

### **March 2022 Examination**

## **INDICATIVE SOLUTION**

#### **Introduction**

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

**Solution 1:**

- i) A surety insurance guarantees another party's performance to a third party in case of default.

A surety bond is provided by the Surety (insurance company) on behalf of the Principal (contractor) to the entity (obligee), which is awarding the project.

When a principal breaks a bond's terms, the harmed party can make a claim on the bond to recover losses.

It can effectively replace the system of bank guarantee, issued by banks for projects, and help reduce risks due to cost overrun, project delays and poor contract performance

[3]

- ii)

1. The Contract

Insurance: Two-party contract between the insured and the insurer.

Surety bond: A contract among at least 3 parties.

[2]

2. Protection

Insurance: Insured is indemnified by the Insurer in case of a loss.

Surety Bond: Protects the obligee if the second party (the principal) defaults.

[2]

3. The Premium

Insurance: Premium is calculated based on expected losses that may happen.

Surety Bond: The premium is generally paid for the guarantee required for the principal to fulfil his obligation.

[2]

4. Losses

Insurance: Losses are expected. The insurance company knows this and pools the risk with the law of large numbers so that the risk is shared.

Surety Bond: Surety bonds are not expected to incur losses. They are issued only to qualified individuals or businesses whose projects require a guarantee. Losses are rare and it is expected that these may be paid back by the principals.

[2]

5. Claims

Insurance: When a claim is paid the insurance company usually doesn't expect to be repaid by the insured. Once claim is settled, salvage is property of Insurer.

Surety Bond: A surety bond is a form of credit, so the principal may be responsible to pay any claims.

[2]

[Max 8 Marks]

[11 Marks]

**Solution 2:**

There will be reduction on the asset value mainly due to:

- Low interest rates for both government and corporate bonds which reduces future income and increases reinvestment risk as proceeds are invested at lower rate; [1]
- Downgrading bonds drives the need for many insurers to rebalance their portfolios into higher rated bonds, thereby realizing and locking in losses; [1]
- Credit and credit spread risk increases risk of defaults and reducing asset values; [1]
- Default of reinsurer; [1]
- Equity market volatility [1]

The movements in interest rates and spreads, in particular, drive changes in the liability discount rates used in many regimes, if liabilities are discounted. These reductions lead to increases in the valuations of the liabilities and a lengthening of their durations.

[2]

[Max 5 Marks]

**Solution 3:**

i) The key objectives of legislation are

- to promote efficient, fair, safe and stable insurance markets
- to help growth and competition in the insurance sector and contribute to economic growth
- enhance the overall efficiency of the financial system
- reduce transaction costs
- create liquidity
- facilitate economies of scale in investment
- allocate resources efficiently
- manage risk
- mobilise long-term savings.

[0.5 Mark for each point, Max 4]

ii) A few of measures taken by Regulators of various countries to protect policyholders are;

- Making statements or provided guidance on the types of COVID-19- related losses that various types of insurance policies will cover. This helped in avoiding disputes and litigations.
- Increased grace period on premium payment.
- Exempting face-to-face meetings to reduce the risk to intermediaries and their clients.
- Exempting dispatching hard copy of policies to policyholder and allowed soft copy can be emailed
- Prioritizing on faster settlement specifically on Covid claims and ensuring that insurers have sufficient liquidity.
- Providing policyholders with various types of flexible premium payment arrangements:
- Health insurance coverage has been adapted to allow for reimbursement of costs for COVID-19 tests and hospitalisation as well as for telemedicine (which may only be covered in limited circumstances under some health insurance plans).
- Extending additional living expenses benefits for residential policyholders whose home repairs have been delayed

[1 Mark for each point, Max 5]

[9 Marks]

**Solution 4:**

Assumptions:

- Premium is written equally throughout the year
- Commission and Expense ratio is calculated on Net Premium
- Loss ratio is provided on Earned Premium
- MDP assumed is 85%
- Dividend Income is calculated on average asset value
- Company portfolio moves in tandem with stock market
- It is assumed Fair value is realised both in fall and increase in Equity portfolio.
- Tax assumed is Nil

[0.5 marks for each point, Max 2]

| Reinsurance Working |       |          |                                  |       |         |
|---------------------|-------|----------|----------------------------------|-------|---------|
| FY                  | EGNPI | ADJ Rate | Minimum Deposit<br>Premium @ 85% | AGNPI | XL Cost |
| 2019-20             | 750   | 2%       | 12.75                            | 500   | 12.75   |
| 2020-21             | 750   | 2%       | 12.75                            | 700   | 14      |

Minimum Deposit premium = EGNPI\*ADJ rate \* 85%

[2]

| Investment Account |                               |                             |                        |                                           |                 |
|--------------------|-------------------------------|-----------------------------|------------------------|-------------------------------------------|-----------------|
| FY                 | Equity value at Start of year | Equity value at End of year | Fair Value realisation | Equity value after fair value realization | Dividend Income |
| 2019-20            | 5,000.0                       | 5,500.0                     | 50.0                   | 5,450.0                                   | 157.5           |
| 2020-21            | 5,450.0                       | 4,905.0                     | -54.5                  | 4,959.5                                   | 155.3           |

[2]

| Revenue Account     |            |            |
|---------------------|------------|------------|
|                     | FY 2019-20 | FY 2020-21 |
| GWP                 | 500.00     | 700.00     |
| Premium Ceded       | 12.75      | 14.00      |
| NWP                 | 487.25     | 686.00     |
| UPR b/f             | 243.63     | 243.63     |
| UPR c/f             | 243.63     | 343.00     |
| NEP                 | 487.25     | 586.63     |
| NIC                 | 414.16     | 498.63     |
| Commission          | 24.36      | 34.30      |
| Expenses            | 73.09      | 102.90     |
| Underwriting Result | -24.36     | -49.21     |
| Investment Income   | 207.50     | 100.83     |
| Insurance Result    | 183.14     | 51.62      |

[4]

| Free Reserve              |            |            |
|---------------------------|------------|------------|
|                           | FY 2019-20 | FY 2020-21 |
| Free Reserve at Start     | 1,000.00   | 1,633.14   |
| Insurance Result          | 183.14     | 51.62      |
| Unrealized gain or losses | 450.00     | -490.50    |
| Free Reserve at End       | 1,633.14   | 1,194.26   |

[2]

[12 Marks]

**Solution 5:**

- The report should compare all actual metrics with pricing assumptions or a benchmark or Industry figures or budgeted figures.
- Motor policy consists of 2 constituents, Motor OD and Motor TP. Both need to be separately analysed.
- The motor TP premium rates are tariff. However, the TP experience across various segments affects the loss ratio and combined ratio for the comprehensive policy.

[2]

The Report should cover:

## 1. Data:

- As the company is large, it will have significant amount of data;
- Source of data and its period, its preparation and verification
- data errors and their resolution
- Perils, coverages which have been captured in the data and used for analysis

[1]

## 2. Analyses of experience may cover:

[max 5]

- Frequency-severity vs burning cost
- Accident year vs underwriting year
- Claims reserves estimation

[0.5]

[0.5]

For the motor TP portion of the product, reserves will be a significant part of the claim experience. As company is large, the company may have its own credible claim delay pattern which may be used for developing the claims

[1]

- Exposure

Business mix by geography, distribution channel, sub segment, age, types of risk, sum insured, new vs renewal vs roll over and other rating factors [1]

- Expense analysis and allocation

Fixed / variable and new / renewal, Direct / Indirect  
Actual expenses should be compared with planned [1]

- Distribution cost [1 mark]

Actual vs Expected - By channels separately for OD and TP  
incentive paid to each channel

- Renewal Rates
- Cancellation rates and reasons
- Grievances
- Any adjustments to data such as IBNR, inflation, and Trending

The main claims analyses will cover: [0.5 mark for each point, Max 3 marks]

- Changing frequency and severity of claims
- Small claims Vs Large claims
- Impact and incidence of large claims
- Catastrophic events and their impact
- Assessing concentration of claims and risk
- Estimation of claim trends

Reinsurance investigations are to analyse the: [0.5 marks for each point, Max 3 marks]

- recoveries on gross claims
- extent of accumulations
- need for catastrophe reinsurance
- value for money
- appropriateness of cover
- profitability of layers
- effects on capital.

Others: [1]

- Actual investment income over the period of the product should be compared with pricing assumption
- Expected capital requirement Vs Actual Capital
- Return on Capital

[15 Marks]

#### **Solution 6:**

We define operational risk as the risk of loss resulting from inadequate or failed internal processes, people and systems or from external events.

[1]

#### 1. Administration Risk

a. The risk of error or failure associated with the administrative aspects of the firm's operations.

b. This risk will depend on the extent to which the firm's administration is outsourced, the extent of centralised and decentralised functions and the level of staff expertise.

c. Examples:

- i. Failure of the third-party IT software vendor.
- ii. Error in claims processing leading to double payments

2. Regulatory / Compliance Risk

- a. The risk of non-adherence to legislative and internal firm requirements.
- b. We might also consider the cost of implementing future regulatory reforms.

c. Examples:

- i. IRDA fines for delayed submission
- ii. Changes in solvency capital calculation methodology, leading to consultancy expenses and increased staff costs.

3. Event Risk

- a. The risk associated with the potential impact of significant events on the operations of the firm.

b. Event risk is not intended to cover the firm's underwriting losses from such events as this would normally be covered within insurance risk.

c. Examples:

- i. a financial system crisis
- ii. a major change in the fiscal system
- iii. natural disaster, for instance, flooding in the city blocking access to the company's headquarter office

4. Fraud Risk

- a. The risk associated with intentional misappropriation of funds, undertaken with the objective of personal benefit at the expense of the firm.

b. Considerations include: the possibility of fraudulent acts occurring within the firm, and the extent of controls which management have established to mitigate this risk

c. Examples:

- i. Embezzlement of funds by an employee
- ii. Accounting fraud by a portfolio manager to show higher than actual profitability of his portfolio

5. Governance Risk

- a. The risk associated with the board and/or senior management of the firm not effectively performing their respective roles

b. Examples:

- i. Not having a risk policy in place, resulting in acceptance of high risk by a new underwriter.
- ii. Not able to raise sufficient capital in a timely manner, required to improve solvency ratio.

6. Strategic Risk

- a. The risk arising from the inability to implement appropriate business plans and strategies, make decisions, allocate resources or adapt to changes in the business environment

b. Examples:

- i. Not adopting to the technological advancements.
- ii. Failure of marketing strategy

iii. Inappropriate reinsurance strategy / structure.

#### 7. Technological Risk

a. The risk of error or failure associated with the technological aspects of its operations.

b. This risk refers to both the hardware systems and the software utilised to run those systems.

c. Examples:

i. Failure of central server

ii. Failure of the claims reporting software due to bug in the code.

#### 8. Cyber risk

a. any risk of financial loss, disruption or damage to the reputation of an organisation from some sort of failure of its information technology systems (can be included with Technological Risk as well)

b. Examples:

i. Company info website hacked – not able to sell new policies

ii. Brute force access to company's database leading to insured's personal information leaked online

#### 9. Pension Scheme Risk

a. The risk that the firm is required to make good any shortfall in pension scheme assets relative to its liabilities.

b. Considerations include:

i. the extent to which extreme events might impact upon the pension scheme funding position

ii. the financial consequences for the firm in such cases

iii. any commitment to fund or eliminate any existing pension scheme deficit. For example, we may consider holding capital to the value of five years' worth of contributions in excess of the standard contributions required to fund the deficit of the pension plan.

c. Examples:

Sudden mass resignation leading to a shortfall in Gratuity fund

Change in calculation methodology for a government-mandated Pension scheme resulting in a sudden increase in the provisions liability.

[18 Marks]

#### Solution 7:

i) It implies that the rows in the triangle are based on the sale date of the policy, whereas in Accident Year cohorts the rows are based on the date the accident (/event) occurred [1]

ii) a) It will include UPR + IBNR if  $LR < 1$ ,  
it will include UPR+PDR+IBNR if  $LR > 1$

b) It will include UPR + OCR + IBNR if  $LR < 1$ ,  
it will include UPR+PDR+OCR + IBNR if  $LR > 1$  [2]

iii) The latest diagonal in the triangle has a much faster/worse development.

Possible reasons:

- i. Data error
- ii. Backlog of payments were cleared
- iii. Late reporting (intermediaries, etc)
- iv. Change in claims handling procedures
- v. Catastrophe event
- vi. Seasonality trend
- vii. Change in consumer behaviour / propensity to claim

[6]

- viii. Court awards  
 ix. Inflation / Increase in repair costs (for instance, in MOD)

(iv) &(v)

Approach 1: Assuming that this was a one-off spike and using BF method:

|        | 1    | 2    | 3    | 4    | 5   | Latest | Initial Ult | BF Ult | Reserves |
|--------|------|------|------|------|-----|--------|-------------|--------|----------|
| 2020Q1 | 400  | 480  | 528  | 570  | 600 | 600    | 600         | 600    | 0        |
| 2020Q2 | 500  | 600  | 660  | 1200 |     | 1200   | 750         | 1237.5 | 37.5     |
| 2020Q3 | 600  | 720  | 1400 |      |     | 1400   | 900         | 1508   | 108      |
| 2020Q4 | 720  | 1500 |      |      |     | 1500   | 1080        | 1716   | 216      |
| 2021Q1 | 1000 |      |      |      |     | 1000   | 1296        | 1296   | 296      |

Approach 2: Assuming that this was a data error and recreating the corrected triangle:

|        | 1   | 2   | 3   | 4     | 5   | Latest | CLUlt | Reserves |
|--------|-----|-----|-----|-------|-----|--------|-------|----------|
| 2020Q1 | 400 | 480 | 528 | 570   | 600 | 600    | 600   | 0        |
| 2020Q2 | 500 | 600 | 660 | 712.5 |     | 712.5  | 750   | 37.5     |
| 2020Q3 | 600 | 720 | 792 |       |     | 792    | 900   | 108      |
| 2020Q4 | 720 | 864 |     |       |     | 864    | 1080  | 216      |
| 2021Q1 | 864 |     |     |       |     | 864    | 1296  | 432      |

[1+8]

vi)

- The claims paid during 2021Q1 calendar period i.e. 2021Q1 diagonal (2021Q1DP11 2020Q4 DP2 and so on..) are much higher than claims paid during other quarters.
- These cells will be outliers in their respective columns.  
 This trend is likely to continue every year, in the sense, every Q1 diagonal in the future years will see a similar spike
- We have to be careful while using Chain Ladder method since if this method is used without adjustments.....  
 ..... it can result in over-estimation or under-estimation of reserves depending on which quarter we are reserving in....  
 ..... we can adjust for it by using two sets of development factors, one for Q1 diagonal and the other for non-Q1 diagonal.  
 We can use other methods such as ELR method, or BF method

[4]

vii)

Lower right triangle:

|                      | 1     | 2     | 3       | 4     | 5      | 6     |
|----------------------|-------|-------|---------|-------|--------|-------|
| 2020Q1               |       |       |         |       |        | 600   |
| 2020Q2               |       |       |         |       | 1237.5 | 1,238 |
| 2020Q3               |       |       |         | 1463  | 1,540  | 1,540 |
| 2020Q4               |       |       | 1,586.4 | 1,713 | 1,803  | 1,803 |
| 2021Q1               |       | 1,150 | 1,265   | 1,366 | 1,438  | 1,438 |
| 2021Q2               | 1000  | 1,200 | 1,320   | 2,400 | 2,580  | 2,580 |
| Q1<br>DIAGONAL<br>DF | 2.083 | 1.944 | 1.818   | 1.053 |        |       |
| OTHER DF             | 1.200 | 1.100 | 1.080   | 1.053 | 1      | 1     |

[8]

[30 Marks]

\*\*\*\*\*